

Pratham Arora

+91-9810855594 | pratham3992@gmail.com | linkedin.com/in/pratham3992arora | github.com/prats3992 | pratham-arora.vercel.app

Education

Plaksha University

B.Tech. in Computer Science & AI

Mohali, India | July 2026

CGPA: 8.20 / 10.0

Research & Applied AI Experience

Undergraduate Researcher

Jan. 2026 – Present

Visual Benchmarking of VLMs | Sup. Prof. Pankaj Pansari | Plaksha University

- Built an automated evaluation pipeline benchmarking 6 VLMs (Gemini 3.1 Pro, GPT-5.4, Gemini-Robotics-ER, Claude Opus 4.7, Gemma 4, Qwen 3.5) on 459 images across 5 physical estimation tasks, across 3 independent runs and 3 blur-degradation tiers.
- Identified a misidentification-vs-miscalibration failure dichotomy through top-10 error analysis: GPT-5.4 fails by misidentifying objects; Gemini variants overestimate weights by 1.8–2.5× despite correct identification.

Undergraduate Researcher

June 2023 – Aug. 2023

Sup. Prof. Sandeep Manjanna | Plaksha University

- Developed preprocessing pipeline (contrast adjustment, noise reduction, edge detection) to optimise **Segment Anything Model (SAM)** for agricultural crop-weed segmentation on sparse datasets; improved batch throughput for **10,000+ images** via vectorized NumPy and multiprocessing.

Industry Experience

AI Engineering Intern

June 2025 – July 2025

Cotality | Kolkata

- Cut AI embedding costs by **85%** by building a hash-based system that fingerprints each function's AST and skips re-embedding unchanged code, eliminating redundant Azure OpenAI API calls.
- Built a production **RAG pipeline** using LangChain and Azure OpenAI, backed by **Cosmos DB with IVF vector indexing** for semantic code retrieval across 15+ language codebases.
- Architected a FastAPI backend with multi-language AST parsing, generating structured documentation embeddings for a unified vector store serving **17+ enterprise repositories**.

Technical Projects

VR LLM Conversational Agent | Unity, Gemini API, Google Cloud STT/TTS

Aug. 2025 – Dec. 2025

- Engineered a real-time VR conversational agent integrating **Gemini 2.5 Flash** with Google Cloud STT/TTS, achieving average response latency of **1.8s** through parallel API calls, caching, and audio preprocessing.
- Designed **within-subjects user study** ($N = 18$) evaluating three latency feedback mechanisms; proved via Wilcoxon test ($p < 0.05$) that embodied gestures outperform visual indicators; **co-authored paper under review at MIT's Presence**.

Kelp Forest Semantic Segmentation | PyTorch, U-Net

Jan. 2024 – May 2024

- Achieved **Top 100 finish (top 17%)** in a DrivenData satellite imagery competition using **U-Net with EfficientNet-B3** backbone for kelp forest segmentation in noisy real-world imagery.
- Built a streak-detection pipeline using an ensemble of **CNN classifiers** (ResNet, DenseNet) with hard voting to filter corrupted training images, achieving **F1 score of 0.952**.

Leadership & Involvement

Head of Technology

Apr. 2025 – Feb. 2026

Athleda – Plaksha Sports Society | Plaksha University

- Built and maintained a sports platform serving **500+ students**, shipping event registration, live-scoring, and photo galleries, while mentoring 2 junior developers through code reviews and production deployments.

LEAP.AI @ Plaksha

Feb. 2024 – Feb. 2025

Web Developer → Head of Web Development | Plaksha University

- Led a team of 7–8 through **2 full platform redesigns** as Head of Web Development, growing the AI club site to **800+ monthly users** for events, workshops, and project showcases.

Skills

AI/ML: PyTorch, LangChain, RAG Pipelines, Segment Anything (SAM), OpenCV

Languages: Python, SQL, JavaScript, TypeScript

Cloud & Data: Azure (Cosmos DB, OpenAI), Firebase, FAISS, Google Cloud

Engineering: FastAPI, Node.js, Next.js, REST APIs, Git/GitHub, Vercel